

Aritra Sadhukhan

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WORK EXPERIENCE

Engineer Web — Codeclouds

Apr 2024 – Present

- Architected a headless, Python-based workflow orchestration engine for a telehealth automation platform, enabling user-defined custom automation templates — including abandoned-cart retargeting, prescription renewal reminders, and admin-configured webhooks — with durable wait states and idempotent, Kafka-driven execution.
- Designed a centralized event logging and traceability pipeline spanning 5 macroservices for a telehealth platform using AWS Firehose, Lambda, S3, and Athena, with Parquet compression and automated lifecycle tiering to reduce long-term storage costs.
- Re-architected a failing product synchronization pipeline handling up to 100,000 products using AWS SQS and Lambda, replacing in-process threaded imports with resilient cursor-based batching to eliminate rate-limit failures and database overload.
- Designed a TypeScript backend framework for Fastify and Express.js adopted across 30+ macroservices, standardizing auth, validation, and Kafka event streaming through a CLI code generator producing functional CRUD modules, cutting new-service boilerplate by ~40%.
- Led load testing and performance optimization for a high-traffic e-commerce checkout platform using Locust, resolving bottlenecks via static routing, Redis/in-memory caching, database indexing, and PM2 clustering to sustain ~30,000 requests/second.
- Engineered a two-tier caching architecture using Redis, in-memory LRU caches, Pub/Sub invalidation, and AWS sticky sessions, improving template rendering throughput by ~5x across horizontally scaled instances.
- Built a multi-tenant analytics backend using Kafka and MongoDB aggregation pipelines, processing ~1M events and 100,000 orders to power sales, conversion, and fraud-detection dashboards refreshed with a 1-day data lag.
- Built a multi-tenant SSO and identity platform using OAuth 2.0/OIDC and JWT-based authentication, supporting 4 MFA methods (TOTP, OTP, passkeys, recovery tokens) with device-level session management and RBAC across internal workspace applications.
- Built an experimentation engine for checkout upsell optimization, enabling configurable A/B and split testing with session-persistent variant assignment across ~1M daily checkout requests.
- Designed an extensible integration framework unifying 5 CRM and e-commerce platforms (Sticky, CheckoutChamp, Vrio, Shopify, WooCommerce) through a shared abstraction layer using the Abstract Factory pattern.
- Designed a custom HTML templating language and synchronous rendering engine for a browser-based checkout builder, serving ~1M requests per day across merchant-customized checkout pages resolved live from Amazon S3.

Full Stack Developer — Codeclouds

Jun 2022 – Apr 2024

- Architected a modular leave accrual engine for an HR management platform, consolidating 52 policy combinations into 4 extensible strategies with cron-driven, idempotent asynchronous processing via Google Cloud Tasks.
- Built and maintained REST APIs and macroservices across 6 core HR platform modules — leave management, ticketing, notifications, scheduling, knowledge base, and centralized logging — using Node.js, Express.js, TypeScript, and Kafka.

LANGUAGES & TECHNOLOGIES

Languages	TypeScript (Node.js), Python, Go, SQL, HTML, CSS
Frameworks & Libraries	Express.js, Fastify, FastAPI, TypeORM, Fiber (Go), Socket.IO, NumPy, pandas, scikit-learn, LangChain, LangGraph
Databases	PostgreSQL, MongoDB, DocumentDB, ChromaDB
Cloud & Infrastructure	AWS (S3, Lambda, SQS, Firehose, Athena, ECS, EC2, IAM, ECR), Docker, Kubernetes, Kafka, Redis, Nginx, Linux, GitHub Actions
Architecture & Protocols	Microservices, Event-Driven Architecture, RAG, REST APIs, RBAC, OAuth 2.0 / OIDC, WebRTC, WebSocket

EDUCATION

B.Tech, Computer Science & Engineering — Ideal Institute of Engineering

Jul 2023 – Jul 2026

CGPA: 8.2

Diploma, Electrical Engineering — North Calcutta Polytechnic

Jul 2018 – Aug 2021

CGPA: 9.1

EXTRACURRICULAR

Projects

Hush — hush.swiftgeek.in

End-to-end encrypted chat application with zero-knowledge architecture, AES-GCM/CTR message and media encryption, RSA-based key exchange, and WebRTC audio/video, with a horizontally scalable Socket.IO backend for chunked, encrypted file transfer.

Certifications

[TypeScript](#) (2024) · [Microservices with Go and gRPC](#) (2025) · [Python](#) (2024) · [Docker](#) (2024) · [MySQL](#) (2025) · [AWS Cloud Practitioner](#) (2026)

Open Source

Contributor to [Twenty](#), [OpenBB](#), [Unkey](#), [Dub](#), [Papermark](#), [Openplane](#), and [Hanko](#). Hacktoberfest: Top-12 finisher (2024), [Super Contributor](#) (2025).

Awards

Outstanding Performance Award In Development — Spectrum (2023).